General Relativity: The Principle of Equivalence and Its Consequences

PHYS 2203 — Introductory Physics III

October 14, 2026

Overview

Special relativity begins with the postulate that the speed of light is the same in all inertial frames. General relativity begins with a different postulate: inertial mass and gravitational mass are the same. This lecture shows how that equality leads to the principle of equivalence, which says that a freely falling frame behaves exactly like an inertial frame without gravity. From it we derive the gravitational redshift and gravitational time dilation and apply them to the Global Positioning System. We then see qualitatively why gravity bends light.

1 Inertial Mass and Gravitational Mass

Mass appears in two different laws of mechanics. Near Earth's surface, Newton's law of gravitation gives the gravitational force on an object, and Newton's second law gives its acceleration:

$$F_{\text{grav}} = \frac{GM_{\text{Earth}}}{r_{\text{Earth}}^2}\, m_g \approx (9.8\ \text{m/s}^2)\, m_g, \qquad a = \frac{F_{\text{net}}}{m_i}. \tag{1}$$

These describe two different properties. The **gravitational mass** m_g determines how hard gravity pulls on an object. The **inertial mass** m_i determines how strongly the object resists being accelerated. There is no obvious reason they should be equal. For a dropped object the net force is the gravitational force, so

$$a = (9.8\ \text{m/s}^2)\,\frac{m_g}{m_i}. \tag{2}$$

If the ratio m_g/m_i differed from one object to another, different objects would fall at different rates. Experiments show that $m_g = m_i$ to better than 1 part in 10^{12}, so all objects fall with the same acceleration.

Gravity and acceleration are indistinguishable

Bob stands in a closet at rest on Earth (frame B), where gravity points down with $g = 9.8\ \text{m/s}^2$. Anna stands in an identical closet far out in space, with no gravity, but a rocket accelerates her closet upward at $9.8\ \text{m/s}^2$ (frame A).



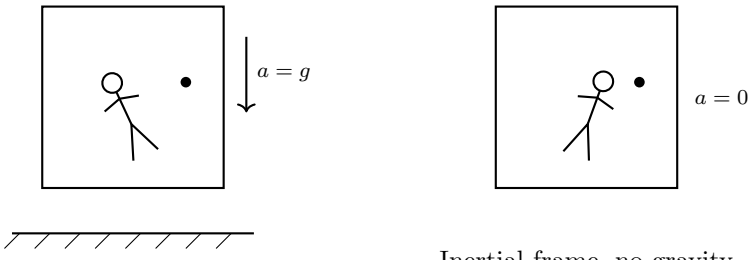
Frame B: Bob at rest on Earth

Frame A: Anna accelerating in gravity-free space

Figure 1: For Bob to remain at rest on Earth, the floor must push up on him with $F_N = m_g g$. For Anna to accelerate with her closet, the floor must push up on her with $F_N = m_i g$. If $m_g = m_i$, the two forces are equal.

To keep Bob at rest, the floor must push up on him with $F_N = m_g g$. To accelerate Anna along with her closet, the floor must push up on her with $F_N = m_i g$. Since $m_g = m_i$, the forces are equal. In fact, no mechanical experiment can tell the two frames apart. In Anna's frame, every object seems to feel an "inertial force" $-m_i \mathbf{a}$, opposite to the acceleration. A dropped object falls to the floor, for example, because once released it no longer accelerates with the closet. Because $m_i = m_g$, this inertial force mimics the gravitational force $m_g \mathbf{g}$ exactly.

The same reasoning applies to a frame falling freely in a uniform gravitational field. The gravitational force $m_g \mathbf{g}$ is exactly cancelled by the inertial force $-m_i \mathbf{a}$, so objects float, just as they do in an inertial frame with no gravity.



Freely falling frame

Inertial frame, no gravity (engine off)

Figure 2: Objects float in a frame falling freely under gravity, just as they do in an inertial frame with no gravity.

2 The Principle of Equivalence

Einstein postulated that *all* physical phenomena, not just mechanical ones, occur identically in a freely falling frame and in an inertial frame without gravity. No experiment of any kind, including experiments with light, could distinguish them.

This generalizes the idea of an inertial frame. A **locally inertial frame** is one falling freely in a gravitational field. It must be small enough ("local") that the field is uniform across it, so that every object in it has the same acceleration. An inertial frame with no gravity is a special case. The fundamental postulate of general relativity is then

The principle of equivalence. The form of each physical law is the same in all locally inertial frames.

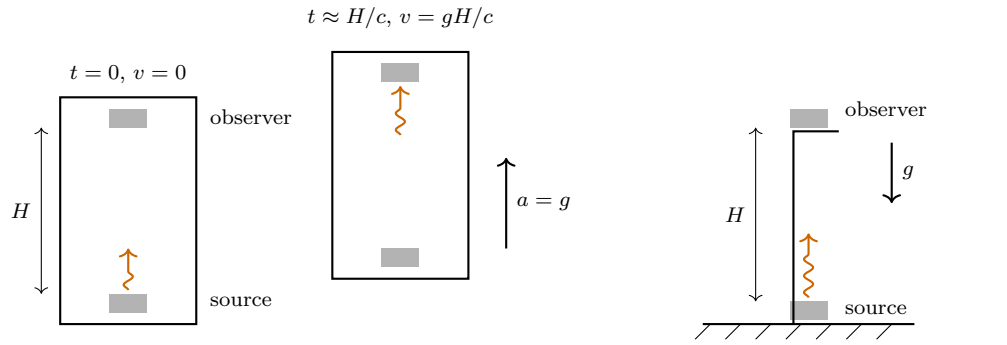
Einstein had this idea in 1907 while working at the patent office in Bern. He realized that a person falling freely would not feel their own weight, and he later called it "the happiest thought of my life."

The full theory that follows from this postulate uses tensors and differential geometry and is beyond this course (an optional sketch is given in the Appendix). Several of its predictions can still be understood from the principle of equivalence alone. Three are especially well known:

1. the gravitational redshift,
2. the deflection of light by the Sun,
3. the precession of the perihelion of Mercury.

3 Gravitational Redshift and Time Dilation

The principle of equivalence predicts that light emitted at one point in a gravitational field has a different frequency when observed at another point. To see this, we first analyze the equivalent situation without gravity: a source and an observer in a frame accelerating at g.



(a) Accelerating frame, no gravity (b) Equivalent frame with gravity

Figure 3: (a) A wave front leaves the source when the accelerating frame is momentarily at rest. By the time it reaches the observer, a time H/c later, the observer is moving away at $v = gH/c$, so there is a Doppler shift. (b) By the principle of equivalence, the same frequency shift occurs between a source and an observer at rest in a gravitational field.

The accelerating frame. The source emits a wave front when the frame is momentarily at rest. The observer, a height H above, receives it about H/c later. By then the frame has gained speed $v = gH/c$, so the observer is moving away from where the light was emitted and sees it Doppler shifted. Using the Doppler formula of Lecture 15 for a receding observer, and the binomial approximation $(1+x)^n \approx 1 + nx$ for small v/c,

$$f_{\text{obs}} = f_{\text{source}} \left(1 - \frac{v}{c}\right)^{1/2} \left(1 + \frac{v}{c}\right)^{-1/2} \approx f_{\text{source}} \left(1 - \frac{1}{2}\frac{v}{c}\right)\left(1 - \frac{1}{2}\frac{v}{c}\right) \approx f_{\text{source}} \left(1 - \frac{v}{c}\right), \qquad (3)$$

where we dropped the small v^2/c^2 term. With $v = gH/c$,

$$\boxed{f_{\text{obs}} \approx f_{\text{source}} \left(1 - \frac{gH}{c^2}\right).} \tag{4}$$

The equivalent frame with gravity. By the principle of equivalence, the same result holds for a source and an observer at rest in a gravitational field g. As light rises in a gravitational field, its frequency decreases and its wavelength increases. This is the **gravitational redshift**.

Gravitational time dilation

The redshift is really a time-dilation effect, even though the source and the observer are not moving relative to each other. Light of wavelength 600 nm has frequency 5×10^{14} Hz, so in one second at the source 5×10^{14} wave fronts are emitted. The observer above receives fewer wave fronts per second, so to see all 5×10^{14} of them the observer must wait *more* than one second. Relative to the observer, the source's clock runs slow:

$$\boxed{\Delta t_{\text{lower}} \approx \Delta t_{\text{higher}} \left(1 - \frac{gH}{c^2}\right).} \tag{5}$$

Clocks deeper in a gravitational field run more slowly. Most fields are weak, with $gH/c^2 \ll 1$, so the effect is small, but modern atomic clocks are precise enough to measure it.

4 Relativity and the Global Positioning System

GPS locates objects on Earth to within a few meters, which requires very precise timekeeping. Each satellite sends a time signal, and an error of only 10 ns corresponds to a distance error of $(3 \times 10^8 \text{ m/s})(10 \times 10^{-9} \text{ s}) = 3$ m. Two relativistic effects compete:

- **Special relativity:** the satellite moves relative to Earth, so its clock runs *slow*.
- **General relativity:** the satellite is higher in Earth's gravitational field, so its clock runs *fast* compared with a clock on the ground.

Equation (5) assumes a uniform field. Earth's field weakens with distance, and the corresponding result is

$$\Delta t_{\text{Earth}} \approx \Delta t_{\text{satellite}} \left[1 - \frac{1}{c^2}\left(\frac{GM}{r_{\text{Earth}}} - \frac{GM}{r_{\text{satellite}}}\right)\right], \tag{6}$$

where M is Earth's mass and r is measured from Earth's *center*. It has the same form as Eq. (5). In both, the correction is the change in gravitational potential energy per unit mass, gH for a uniform field and $GM/r_{\text{Earth}} - GM/r_{\text{satellite}}$ for Earth's field.

Example: correcting GPS clocks

A GPS satellite orbits at an altitude of 2.0×10^7 *m with a speed of* 3.9×10^3 *m/s. Earth's radius is* 6.4×10^6 *m. (a) By approximately what fraction must the satellite's time be adjusted to account for both effects? (b) If this were not done, how soon would the time be in error by* 10 *ns?*

(a) For special relativity, the satellite's speed is much larger than that of a point on Earth's surface, so we use the satellite's speed alone:

$$\Delta t_{\text{Earth}} = \frac{\Delta t_{\text{satellite}}}{\sqrt{1 - \left(\dfrac{3.9 \times 10^3}{3 \times 10^8}\right)^2}} \approx \left(1 + 8.5 \times 10^{-11}\right) \Delta t_{\text{satellite}}. \tag{7}$$

For general relativity, the satellite's distance from Earth's center is $6.4 \times 10^6 + 2.0 \times 10^7 = 2.64 \times 10^7$ m, and $GM = 4.0 \times 10^{14}$ m^3/s^2. Equation (6) gives

$$\Delta t_{\text{Earth}} \approx \Delta t_{\text{satellite}} \left[1 - \frac{4.0 \times 10^{14}}{(3 \times 10^8)^2} \left(\frac{1}{6.4 \times 10^6} - \frac{1}{2.64 \times 10^7}\right)\right] = \left(1 - 5.26 \times 10^{-10}\right) \Delta t_{\text{satellite}}. \tag{8}$$

Adding the two fractional changes,

$$\frac{\Delta t_{\text{Earth}} - \Delta t_{\text{satellite}}}{\Delta t_{\text{satellite}}} \approx -5.26 \times 10^{-10} + 8.5 \times 10^{-11} = -4.4 \times 10^{-10}. \tag{9}$$

The gravitational effect is larger, so overall the satellite's clock runs fast.

(b) The error reaches 10 ns when

$$\frac{10 \times 10^{-9} \text{ s}}{\Delta t_{\text{satellite}}} = 4.4 \times 10^{-10} \qquad \Longrightarrow \qquad \Delta t_{\text{satellite}} \approx 23 \text{ s}. \tag{10}$$

Within a minute the system would be useless. In practice, the satellite clocks are preset to account for the gravitational effect, and receivers on the ground correct for their own motion and elevation.

5 Warped Space-Time

If gravity changes time intervals, we might expect it to change distances as well. A central idea of general relativity is that a massive body **warps space-time** around it. This is easiest to picture in two dimensions, where space is a surface. Inhabitants of the two-dimensional space in Figure 4 expect every grid cell to have the same area. Looking from outside, we can see that cells near the massive body are larger. In the same way, we cannot see the warping of our own three-dimensional space, because we cannot step outside it.

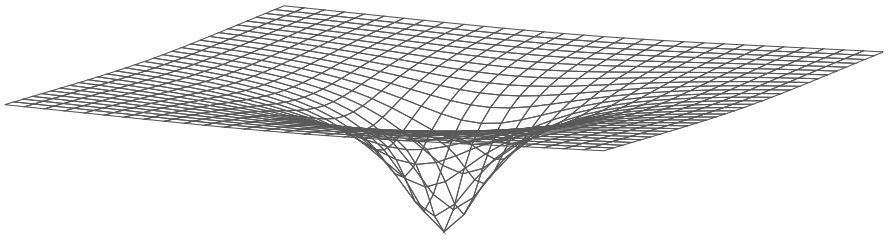
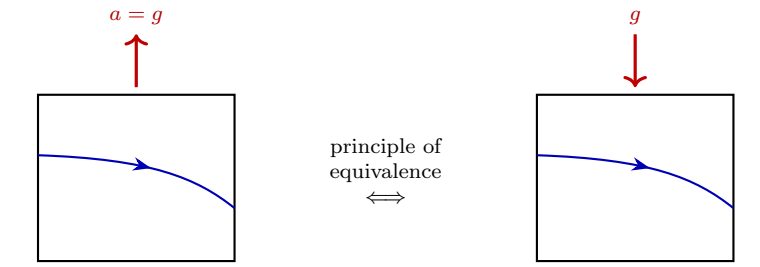


Figure 4: A two-dimensional picture of a massive body warping space. The body sits at the bottom of the depression.

The gravitational redshift shows that **gravity affects light**, even though light has no mass and its energy is all kinetic. Newton's law $F = GMm/r^2$ cannot describe this. It is a special case, correct only when the field is weak, and in that limit the effect on light is negligible. In a strong field, general relativity replaces Newton's law with the picture of warped space-time, and the change in light's frequency is a natural result of passing through a region where space and time are distorted.

6 Deflection of Light by the Sun

If gravity affects light moving directly away from a massive body, light moving sideways should curve. A light beam crossing Anna's accelerating closet appears to curve toward the floor, because the closet accelerates upward while the light crosses it. By the principle of equivalence, the beam must also curve in Bob's closet on Earth.



(a) Accelerating frame without gravity

(b) Equivalent frame with gravity

Figure 5: A light beam crossing a closet. (a) In a closet accelerating upward, the beam appears to curve toward the floor. (b) By the principle of equivalence, the beam must curve the same way in a closet at rest in a gravitational field.

Rather than use Newton's law, we say that light follows the most natural path through warped space-time. A principle from classical optics still applies: *light takes the path of minimum time between two points.* Near a massive body this path is not what an observer with a "flat" view of space would call a straight line. To that observer, light passing near the body appears to curve, and its source appears to be in a different place.

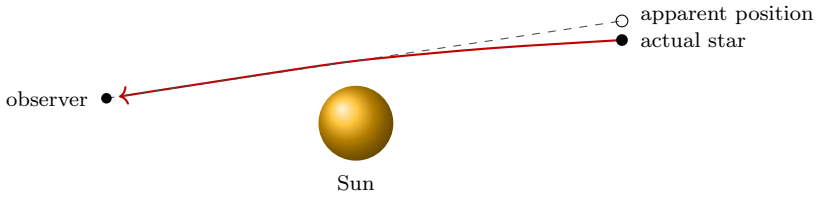


Figure 6: Top view, not to scale. Light from a star bends as it passes the Sun. The observer sees the star along the direction the light arrives from, so its apparent position is shifted away from the Sun.

The Sun warps space only slightly. Its gravitational redshift, for example, is only about 2 parts in 10^6. To see the deflection, we must observe starlight passing very close to the Sun, which is normally drowned out by the Sun's brightness. The observation must therefore be made during a solar eclipse. Observations during the eclipse of 1919 showed a deflection of about 2 seconds of arc for light grazing the Sun, in excellent agreement with general relativity. The bending of light by massive objects in space is called **gravitational lensing**.

A (Optional) The Formalism of General Relativity

This appendix is optional and will not appear on any exam. Everything above was deduced from the equivalence principle alone, without the mathematics of the full theory. What follows is a two-page sketch of what that mathematics looks like, for students who are curious. It builds directly on the **four-vectors** of Lectures 14 and 18.

A.1 From the invariant interval to the metric

Lecture 14 showed that although observers disagree about time separations and distances, they all agree on the *spacetime interval*. For two nearby events,

$$\mathrm{d}s^2 = (c\,\mathrm{d}t)^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2. \tag{11}$$

Label the four coordinates $x^0 = ct$, $x^1 = x$, $x^2 = y$, $x^3 = z$, so that a Greek index μ runs over $0, 1, 2, 3$. The interval can then be written as a double sum,

$$\boxed{\mathrm{d}s^2 = \sum_{\mu=0}^{3}\sum_{\nu=0}^{3} \eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu,} \qquad \eta_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}, \tag{12}$$

where $\eta_{\mu\nu}$ is the **Minkowski metric**. The metric is the bookkeeping device that tells you how to build an invariant. For *any* four-vector $A^\mu = (A_t, A_x, A_y, A_z)$,

$$\sum_{\mu,\nu} \eta_{\mu\nu} A^\mu A^\nu = A_t^2 - \left(A_x^2 + A_y^2 + A_z^2\right), \tag{13}$$

which is exactly the invariant of Lecture 14. Applied to the position-time four-vector it gives the interval. Applied to the energy–momentum four-vector $(E/c,\, p_x,\, p_y,\, p_z)$ of Lecture 18 it gives m^2c^2, the mass.

A.2 Curvature: a metric that varies from place to place

Here is the single central idea of general relativity. In the presence of gravity, spacetime is **curved**, and the constant array $\eta_{\mu\nu}$ is replaced by a **metric tensor** $g_{\mu\nu}(x)$ whose components vary from event to event:

$$\boxed{\mathrm{d}s^2 = \sum_{\mu,\nu} g_{\mu\nu}(x)\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu.} \tag{14}$$

Being symmetric, $g_{\mu\nu}$ has 10 independent components — ten functions of position that encode the gravitational field entirely.

The *principle of equivalence* of Section 2 has a precise statement in this language: at any single event one can always choose coordinates — those of a freely falling observer — in which $g_{\mu\nu} = \eta_{\mu\nu}$ and its first derivatives vanish. That is exactly what "locally inertial" means, and it is why special relativity always holds locally. What *cannot* be transformed away are the *second* derivatives of $g_{\mu\nu}$. Those encode genuine curvature, observable as tidal effects.

A.3 Geodesics: how matter moves

In general relativity there is no gravitational *force*. A freely falling particle simply follows a **geodesic** — the straightest available worldline, the path that extremizes the elapsed proper time $\int \mathrm{d}\tau$. This generalizes the closing remark of Lecture 14: in flat spacetime the straight worldline between two events carries the *greatest* proper time, which is why the stay-at-home twin ages more. The same principle governs curved spacetime, except that the favored path is no longer straight in the everyday sense. Planets orbit the Sun because they are following the straightest available paths through a curved geometry.

The geodesic takes the form

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2} + \sum_{\alpha,\beta} \Gamma^\mu_{\alpha\beta} \frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau} \frac{\mathrm{d}x^\beta}{\mathrm{d}\tau} = 0, \tag{15}$$

where the coefficients $\Gamma^\mu_{\alpha\beta}$ are built from first derivatives of $g_{\mu\nu}$. They play the role of the gravitational field: where the metric varies, $\Gamma \neq 0$ and worldlines bend. The bending of light in Section 6 is this equation applied to a light ray.

A.4 The stress–energy tensor: the source of curvature

What produces the curvature? In Newtonian gravity the source is the mass density ρ. That cannot be the whole answer here, because mass and energy are the same thing (Lecture 18), and energy and momentum are frame-dependent components of a single four-vector.

For one particle, energy and momentum combined into the four-vector $(E/c,\, p_x,\, p_y,\, p_z)$. For a continuous distribution of matter we must say, for each of those four quantities, how much of it flows in each of the four directions. That requires a 4×4 array, the **stress–energy tensor** $T^{\mu\nu}$:

$$T^{\mu\nu} = \left[\begin{array}{c|c} \text{energy density} & \text{energy flux} \\ \hline \text{momentum density} & \begin{array}{c}\text{pressure and} \\ \text{shear stresses}\end{array} \end{array}\right] \tag{16}$$

Component by component: T^{00} is the energy density, T^{0i} the momentum density (equivalently the energy flux), and T^{ij} the flux of i-momentum in the j direction — the diagonal entries being pressure and the off-diagonal ones shear stress. Like $g_{\mu\nu}$, it is symmetric with 10 independent components. The conservation of energy and momentum, which Lecture 18 showed to be one four-dimensional law rather than two, becomes the single statement $\sum_\nu \partial_\nu T^{\mu\nu} = 0$.

Note that *pressure gravitates*: in relativity all forms of energy, not just rest mass, act as a source of gravity.

A.5 The Einstein field equations

Curvature is measured by the Riemann tensor, assembled from second derivatives of $g_{\mu\nu}$, and contracting it yields the Ricci tensor $R_{\mu\nu}$ and the scalar R. Einstein's field equations then read

$$\boxed{R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}.} \tag{17}$$

The left side is pure geometry, the right side is matter and energy. These are ten coupled nonlinear partial differential equations for the ten functions $g_{\mu\nu}$, and their nonlinearity is what makes general

relativity hard: gravitational energy itself gravitates. John Wheeler's summary is hard to improve on:

> *Spacetime tells matter how to move; matter tells spacetime how to curve.*

The Newtonian limit. For weak fields and slow motion one finds $g_{00} \approx 1 + 2\Phi/c^2$, where Φ is the ordinary gravitational potential, and the field equations collapse to

$$\nabla^2 \Phi = 4\pi G \rho, \tag{18}$$

which is Newton's law of gravity. The pattern is the same one we saw in Lecture 13, where the Lorentz transformation reduced to the Galilean one for $v \ll c$: the newer theory does not discard the older, but contains it as a limiting case.

A.6 Example: the Schwarzschild metric

Solving the field equations for the spacetime outside a spherical mass M, such as a star or planet, gives the **Schwarzschild metric**

$$\mathrm{d}s^2 = \left(1 - \frac{R_s}{r}\right)(c\,\mathrm{d}t)^2 - \frac{\mathrm{d}r^2}{1 - R_s/r} - r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right), \qquad R_s = \frac{2GM}{c^2}. \tag{19}$$

The gravitational redshift of Section 3 can be read straight off. A clock held at rest at radius r has $\mathrm{d}r = \mathrm{d}\theta = \mathrm{d}\phi = 0$, so its own proper time $\mathrm{d}\tau = \mathrm{d}s/c$ obeys

$$\mathrm{d}\tau = \sqrt{1 - \frac{R_s}{r}}\;\mathrm{d}t. \tag{20}$$

Clocks deep in a gravitational well run slow compared with a distant clock. As $r \to R_s$ we get $\mathrm{d}\tau \to 0$: the redshift becomes total, and no light can escape. An object smaller than its own R_s is a *black hole*.